

1.



3. Calculamos NULL, FIRST y FOLLOW

1. Gramática - G

No terminales: $\Delta = \{A', A, B, C\}$ $A \rightarrow @AB \mid @$ terminales: $\Sigma = \{@, \&, m, f, t\}$ $B \rightarrow C \& B \mid Am$ $C \rightarrow f \mid t$ NULL: $\Delta \rightarrow \text{bool}$

2. Gramática Aumentada

0. $A' \rightarrow A$ (Ninguno es nullificable dado que no tenemos transiciones a ϵ)1. $A \rightarrow @AB$

FIRST

2. $A \rightarrow @$ 3. $B \rightarrow C \& B$ 4. $B \rightarrow Am$ 5. $C \rightarrow f$ 6. $C \rightarrow t$

	0	1	2	3
A'	$\emptyset$	$\emptyset$	$\{@\}$	$\{@\}$
A	$\emptyset$	$\{@\}$	$\{@\}$	$\{@\}$
B	$\emptyset$	$\emptyset$	$\{@, f, t\}$	$\{@, f, t\}$
C	$\emptyset$	$\{f, t\}$	$\{f, t\}$	$\{f, t\}$

FOLLOW

	0	1	2	3
A'	$\{\$ \}$	$\{\$, f, t, @, m\}$	$\{\$, f, t, @, m\}$	$\{\$, f, t, @, m\}$
B	$\emptyset$	$\{\$ \}$	$\{\$, f, t, @, m\}$	$\{\$, f, t, @, m\}$
C	$\emptyset$	$\{\& \}$	$\{\& \}$	$\{\& \}$

2.

4. AUTOMATA LR(1)

$$\text{FIRST}(\epsilon \$) = \{\$ \}$$

$$\checkmark I_0 = \text{Closure}([A' \rightarrow \cdot A, \$]) = \{[A' \rightarrow \cdot A, \$], [A \rightarrow \cdot @AB, \$], [A \rightarrow \cdot @, \$]\} \checkmark$$

$$\checkmark I_1 = \text{Goto}(I_0, A) = \{[A' \rightarrow A \cdot, \$]\} \checkmark$$

$$\text{FIRST}(B \$) = \{\$, f, t\}$$

$$\checkmark I_2 = \text{Goto}(I_0, @) = \{[A \rightarrow @ \cdot AB, \$], [A \rightarrow @ \cdot, \$]\} \cup \{[A \rightarrow \cdot @AB, @], [A \rightarrow \cdot @, @], [A \rightarrow \cdot @AB, f], [A \rightarrow \cdot @, f], [A \rightarrow \cdot @AB, t], [A \rightarrow \cdot @, t]\} \cup \{(\cdot @)\}$$

$$\text{FIRST}(\epsilon \$) = \{\$ \} \quad \text{FIRST}(\& B \$) = \{\&\} \quad \text{FIRST}(m \$) = m$$

$$\checkmark I_3 = \text{Goto}(I_2, A) = \{[A \rightarrow @A \cdot B, \$], [B \rightarrow \cdot C \& B, \$], [B \rightarrow \cdot Am, \$], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @AB, m], [A \rightarrow \cdot @, m]\}$$

$$\text{FIRST}(B @) = \text{FIRST}(B f) = \text{FIRST}(B t) = \{\$, f, t\}$$

$$\checkmark I_4 = \text{Goto}(I_2, @) = \{[A \rightarrow @A \cdot B, @], [A \rightarrow @A \cdot, @], [A \rightarrow @A \cdot B, f], [A \rightarrow @A \cdot, f], [A \rightarrow @A \cdot B, t], [A \rightarrow @A \cdot, t]\} \cup \{(\cdot @)\}$$

$$\checkmark I_5 = \text{Goto}(I_3, B) = \{[A \rightarrow @AB \cdot, \$]\}$$

$$\checkmark I_6 = \text{Goto}(I_3, C) = \{[B \rightarrow C \cdot \& B, \$]\}$$

$$\checkmark I_7 = \text{Goto}(I_3, A) = \{[B \rightarrow A \cdot m, \$]\}$$

$$\checkmark I_8 = \text{Goto}(I_3, f) = \{[C \rightarrow f \cdot, \&]\}$$

$$\checkmark I_9 = \text{Goto}(I_3, t) = \{[C \rightarrow t \cdot, \&]\}$$

$$\text{FIRST}(B m) = \{\$, f, t\}$$

$$\checkmark I_{10} = \text{Goto}(I_3, @) = \{[A \rightarrow @A \cdot B, m], [A \rightarrow @A \cdot, m]\} \cup \{(\cdot @)\}$$

$$\checkmark I_{11} = \text{Goto}(I_4, A) = \{[A \rightarrow @A \cdot B, @], [A \rightarrow @A \cdot B, f], [A \rightarrow @A \cdot B, t], [B \rightarrow \cdot C \& B, @], [B \rightarrow \cdot Am, @], [B \rightarrow \cdot C \& B, f], [B \rightarrow \cdot Am, f], [B \rightarrow \cdot C \& B, t], [B \rightarrow \cdot Am, t], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @AB, m], [A \rightarrow \cdot @, m]\}$$

$$\checkmark I_4 = \text{goto}(I_4, @)$$

$$\checkmark I_{12} = \text{goto}(I_6, \&) = \{ [B \rightarrow C \& B, \$], [B \rightarrow \cdot C \& B, \$], [B \rightarrow \cdot A m, \$], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @AB, m], [A \rightarrow \cdot @, m] \}$$

$$\checkmark I_{13} = \text{goto}(I_7, m) = \{ [B \rightarrow A m, \$] \}$$

$$\checkmark I_{14} = \text{goto}(I_{10}, A) = \{ [A \rightarrow @AB, m], [B \rightarrow \cdot C \& B, m], [B \rightarrow \cdot A m, m], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow @AB, m], [A \rightarrow \cdot @, m] \}$$

$$I_4 = \text{goto}(I_{10}, @)$$

$$\checkmark I_{15} = \text{goto}(I_{11}, B) = \{ [A \rightarrow @AB, @], [A \rightarrow @AB, f], [A \rightarrow @AB, t] \}$$

$$\checkmark I_{16} = \text{goto}(I_{11}, C) = \{ [B \rightarrow C \& B, @], [B \rightarrow C \& B, f], [B \rightarrow C \& B, t] \}$$

$$\checkmark I_{17} = \text{goto}(I_{11}, A) = \{ [B \rightarrow A m, @], [B \rightarrow A m, f], [B \rightarrow A m, t] \}$$

$$\checkmark I_8 = \text{goto}(I_{11}, f), I_9 = \text{goto}(I_{11}, t), I_{10} = \text{goto}(I_{11}, @)$$

$$\checkmark I_{18} = \text{goto}(I_{12}, B) = \{ [B \rightarrow C \& B, \$] \}$$

$$\checkmark I_6 = \text{goto}(I_{12}, C), I_7 = \text{goto}(I_{12}, A), I_8 = \text{goto}(I_{12}, f)$$

$$\checkmark I_9 = \text{goto}(I_{12}, t), I_{10} = \text{goto}(I_{12}, @)$$

$$\checkmark I_{19} = \text{goto}(I_{14}, B) = \{ [A \rightarrow @AB, m] \}$$

$$\checkmark I_{20} = \text{goto}(I_{14}, C) = \{ [B \rightarrow C \& B, m] \}$$

$$\checkmark I_{21} = \text{goto}(I_{14}, A) = \{ [B \rightarrow A m, m] \}$$

$$\checkmark I_8 = \text{goto}(I_{14}, f), I_9 = \text{goto}(I_{14}, t), I_{10} = \text{goto}(I_{14}, @)$$

4.

$FIRST(\&Bt) = FIRST(\&B@) = FIRST(\&Bf) = \{\&\}$ (lo mismo con n)



$FIRST(\epsilon @) = \{@\}$, $FIRST(\epsilon f) = \{f\}$, $FIRST(\epsilon t) = \{t\}$

$$\checkmark I_{22} = GOTO(I_{16}, \&) = \{ [B \rightarrow C \& B, @], [B \rightarrow C \& B, f], [B \rightarrow C \& B, t], [B \rightarrow \cdot C \& B, @], [B \rightarrow \cdot C \& B, f], [B \rightarrow \cdot C \& B, t], [B \rightarrow \cdot A m, @], [B \rightarrow \cdot A m, f], [B \rightarrow \cdot A m, t], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @ A B, m], [A \rightarrow \cdot @, m] \}$$

$$\checkmark I_{23} = GOTO(I_{17}, m) = \{ [B \rightarrow A m, @], [B \rightarrow A m, f], [B \rightarrow A m, t] \}$$

$$\checkmark I_{24} = GOTO(I_{20}, \&) = \{ [B \rightarrow C \& B, m], [B \rightarrow \cdot C \& B, m], [B \rightarrow \cdot A m, m], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @ A B, m], [A \rightarrow \cdot @, m] \}$$

$$\checkmark I_{25} = GOTO(I_{21}, m) = \{ [B \rightarrow A m, m] \}$$

$$\checkmark I_{26} = GOTO(I_{22}, B) = \{ [B \rightarrow C \& B, @], [B \rightarrow C \& B, f], [B \rightarrow C \& B, t] \}$$

$$\checkmark I_{16} = GOTO(I_{22}, C), I_{17} = GOTO(I_{22}, A), I_8 = GOTO(I_{22}, f)$$

$$\checkmark I_9 = GOTO(I_{22}, t), I_{10} = GOTO(I_{22}, @)$$

$$\checkmark I_{27} = GOTO(I_{24}, B) = \{ [B \rightarrow C \& B, m] \}$$

$$\checkmark I_{20} = GOTO(I_{24}, C), I_{21} = GOTO(I_{24}, A), I_8 = GOTO(I_{24}, f)$$

$$\checkmark I_9 = GOTO(I_{24}, t), I_{10} = GOTO(I_{24}, @)$$

5.

5. AUTOMATA LALR(1)

$$I'_0 = \text{Closure}([A' \rightarrow \cdot A, \$]) = \{A' \rightarrow \cdot A, \$], [A \rightarrow \cdot @B, \$], [A \rightarrow \cdot @, \$]\}$$

$$I'_1 = \text{Goto}(I'_0, A) = \{[A' \rightarrow A \cdot, \$]\}$$

$$I'_2 = I_2 \cup I_4 \cup I_{10} = \{[A \rightarrow @ \cdot AB, \$/@/f/t/m], [A \rightarrow @ \cdot, \$/@/f/t/m], [A \rightarrow \cdot @AB, @/f/t], [A \rightarrow \cdot @, @, f, t]\} \\ = \text{Goto}(I'_0, @)$$

$$I'_3 = I_3 \cup I_{11} \cup I_{14} = \{[A \rightarrow @A \cdot B, \$/@/f/t/m], [B \rightarrow \cdot C \& B, \$/@/f/t/m], [B \rightarrow \cdot A_m, \$/@/f/t/m], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @AB, m], [A \rightarrow \cdot @, m]\} = \text{Goto}(I'_2, A)$$

$$\text{Goto}(I'_2, @) = I'_2$$

$$I'_4 = I_5 \cup I_{15} \cup I_{19} = \{[A \rightarrow @AB \cdot, \$/@/f/t/m]\} \\ = \text{Goto}(I'_3, B)$$

$$I'_5 = I_6 \cup I_{16} \cup I_{20} = \{[B \rightarrow C \cdot \& B, \$/@/f/t/m]\} \\ = \text{Goto}(I'_3, C)$$

$$I'_6 = I_7 \cup I_{17} \cup I_{21} = \{[B \rightarrow A \cdot m, \$/@/f/t/m]\} = \text{Goto}(I'_3, A)$$

$$I'_7 = \text{Goto}(I'_3, f) = \{[C \rightarrow f \cdot, \&]\}$$

$$I'_8 = \text{Goto}(I'_3, t) = \{[C \rightarrow t \cdot, \&]\}$$

$$\text{Goto}(I'_3, @) = I'_2$$

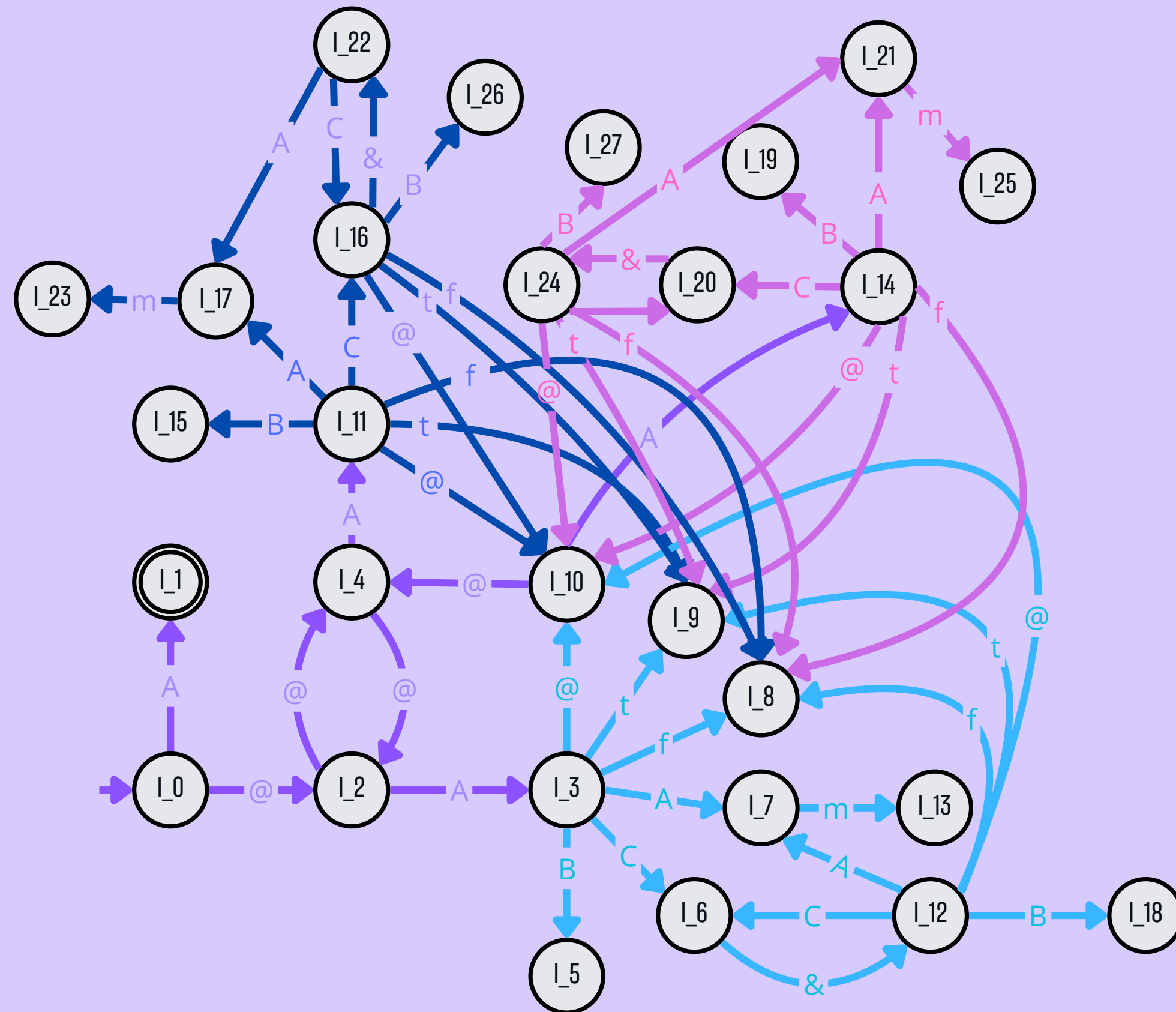
$$I'_9 = \text{Goto}(I'_5, \&) = I_{12} \cup I_{22} \cup I_{24} = \{[B \rightarrow C \& \cdot B, \$/@/f/t/m], [B \rightarrow \cdot C \& B, \$/@/f/t/m], [B \rightarrow \cdot A_m, \$/@/f/t/m], [C \rightarrow \cdot f, \&], [C \rightarrow \cdot t, \&], [A \rightarrow \cdot @AB, m], [A \rightarrow \cdot @, m]\}$$

$$I'_{10} = \text{Goto}(I'_6, m) = \{[B \rightarrow A m \cdot, \$/@/f/t/m]\}$$

$$I'_{11} = \text{Goto}(I'_9, B) = I_{18} \cup I_{26} \cup I_{27} = \{[B \rightarrow C \& B \cdot, \$/@/f/t/m]\}$$

$$\text{Goto}(I'_9, C) = I'_5, \text{Goto}(I'_9, A) = I'_6, \text{Goto}(I'_9, f) = I'_7, \text{Goto}(I'_9, t) = I'_8, \text{Goto}(I'_9, @) = I'_2$$

Autómata LR(1)



Autómata LALR(1)

